CAR SALES ANALYSIS

SQL Portfolio Project

Kabeer Khan | April 2026 | PostgreSQL

■ **Author**	Kabeer Khan	■■ **Database**	PostgreSQL
■ **Date**	April 2026	■ **Dataset**	Car Sales Records
■ **Columns**	13	■ **Questions**	12 Business Queries

■ SECTION 1 — BUSINESS CASE STUDY

Background

AutoDrive Motors is a mid-sized automobile dealership network operating across multiple regions in the United States. The company sells cars from various manufacturers through a network of regional dealers. Despite growing operations, the management team lacks a clear, data-driven view of sales performance, customer behavior, and inventory efficiency.

Business Problem

The company is facing the following challenges:

- Revenue is growing but management cannot identify **which regions, dealers, or models** are driving it.
- **Inventory is misaligned** — some regions are stocking models that do not sell well locally.
- **Customer segments** (by income, gender) are not being targeted with the right products.
- **Monthly sales fluctuate** unpredictably with no early warning system for declining months.

• **Underperforming dealers** are not being identified or trained in time.

Business Objective

The goal of this SQL analysis project is to extract actionable business intelligence from the car sales transaction data to help AutoDrive Motors make smarter decisions about marketing, inventory, dealer management, and customer targeting.

What We Aim to Find & Solve

#	Business Question	Problem to Solve
1	Total Revenue	Understand overall business scale
2	Region-wise Sales	Find top & weak regions
3	Income vs Purchase	Segment customers for targeting
4	Top 3 Models per Region	Optimize regional inventory
5	Month-over-Month Growth	Detect sales decline early
6	Most Popular Model	Focus marketing resources
7	Above Average Dealers	Identify top performers to learn from
8	Body Style Popularity	Stock right vehicle types
9	Transmission Preference	Balance automatic vs manual stock
10	Underperforming Models	Remove or replace poor sellers
11	Gender Purchase Behavior	Gender-targeted campaigns
12	Color vs Body Style	Color-specific inventory decisions

■■ SECTION 2 — DATABASE SCHEMA

```
DROP TABLE IF EXISTS car_sales;

CREATE TABLE car_sales (
car_id VARCHAR(20) PRIMARY KEY,
date DATE,
customer_name VARCHAR(50),
gender VARCHAR(10),
annual_income NUMERIC(12,2),
company VARCHAR(50),
model VARCHAR(50),
transmission VARCHAR(20),
color VARCHAR(20),
price NUMERIC(10,2),
dealer_name VARCHAR(50),
dealer_region VARCHAR(50),
body_style VARCHAR(20)
);
```

Column	Data Type	Description
car_id	VARCHAR(20)	Unique car transaction ID (Primary Key)
date	DATE	Date of sale
customer_name	VARCHAR(50)	Name of the buyer
gender	VARCHAR(10)	Gender of the customer
annual_income	NUMERIC(12,2)	Customer annual income
company	VARCHAR(50)	Car manufacturer brand
model	VARCHAR(50)	Car model name
transmission	VARCHAR(20)	Automatic or Manual
color	VARCHAR(20)	Exterior color of the car
price	NUMERIC(10,2)	Sale price of the car
dealer_name	VARCHAR(50)	Name of the selling dealer
dealer_region	VARCHAR(50)	Geographic region of dealer
body_style	VARCHAR(20)	SUV / Sedan / Hatchback etc.

■ SECTION 3 — SQL QUERIES & ANALYSIS

Q1 What is Total Revenue?

```
SELECT '$ ' || ROUND(SUM(price)/1000000.0, 2) || ' M'
AS total_revenue
FROM car_sales;
```

■ Insight	■ Recommendations
Shows the overall revenue scale of the business.	*• Focus marketing on high-selling models* *• Use as baseline KPI for annual targets*

Q2 Which Region Generates Highest Sales?

```
SELECT dealer_region,
'$ ' || ROUND(SUM(price)/1000000.0, 2) || 'M' AS total_sales
FROM car_sales
GROUP BY dealer_region
ORDER BY SUM(price)/1000000.0 DESC;
```

■ Insight	■ Recommendations
Identifies which geographic regions are the top revenue contributors.	*• Invest more marketing in low-performing regions* *• Expand dealer network in top regions*

Q3 Customer Income vs Purchase Behavior

```
SELECT
CASE
WHEN annual_income < 200000 THEN 'Low Income'
WHEN annual_income BETWEEN 200000 AND 800000 THEN 'Middle Income'
ELSE 'High Income'
END AS income_group,
COUNT(*) AS customers,
ROUND(AVG(price),2) AS avg_purchase
FROM car_sales
GROUP BY income_group;
```

■ Insight	■ Recommendations
Reveals how customer income levels relate to car purchase prices.	*• Target premium cars to high-income group* *• Offer financing options for low-income buyers*

Q4 Top 3 Selling Cars in Each Region

```
WITH top_sales AS (
SELECT dealer_region, model, COUNT(*) AS total_selling,
RANK() OVER(PARTITION BY dealer_region
ORDER BY COUNT(*) DESC) AS rank
FROM car_sales
GROUP BY dealer_region, model
)
SELECT * FROM top_sales WHERE rank <= 3;
```

■ Insight	■ Recommendations
Each region has its own best-selling models — regional preferences differ significantly.	*• Customize regional inventory based on local demand* *• Avoid overstocking non-popular models per region*

Q5 Sales Growth Month-over-Month

```
WITH monthly_sales AS (
SELECT DATE_TRUNC('month', date)::DATE AS month,
SUM(price) AS revenue
FROM car_sales GROUP BY month
)
SELECT month, revenue,
LAG(revenue) OVER(ORDER BY month) AS prev_month,
ROUND(
(revenue - LAG(revenue) OVER(ORDER BY month)) * 100.0 /
NULLIF(LAG(revenue) OVER(ORDER BY month), 0)
, 2) AS growth_percentage
FROM monthly_sales;
```

■ Insight	■ Recommendations
Certain months show revenue decline — sales growth is seasonal and unstable.	*• Run promotions during historically low months* *• Plan seasonal campaigns proactively*

Q6 Most Popular Car Model Overall

```
SELECT company, model, COUNT(*) AS total_sales
FROM car_sales
GROUP BY company, model
ORDER BY total_sales DESC
LIMIT 1;
```

■ Insight	■ Recommendations
The single best-selling model across all regions and dealers.	*• Concentrate marketing budget on the top model* *• Ensure sufficient stock of this model nationwide*

Q7 Which Dealers Perform Above Average?

```
WITH dealer_sales AS (
SELECT dealer_name, SUM(price) AS revenue
FROM car_sales
GROUP BY dealer_name
)
SELECT * FROM dealer_sales
WHERE revenue > (SELECT AVG(revenue) FROM dealer_sales);
```

■ Insight	■ Recommendations
Only a few dealers consistently outperform the network average.	*• Study strategies from top dealers and replicate* *• Provide targeted training for underperforming dealers*

Q8 Which Body Style is Most Popular?

```
SELECT body_style, COUNT(*) AS total_sales
FROM car_sales
GROUP BY body_style
ORDER BY total_sales DESC;
```

■ Insight	■ Recommendations
SUVs and Hatchbacks dominate customer preferences.	*• Increase SUV and Hatchback inventory* *• Reduce stock of less popular body styles*

Q9 Which Transmission Type Do Customers Prefer?

```
SELECT transmission, COUNT(*) AS total_sales
FROM car_sales
GROUP BY transmission;
```

■ Insight	■ Recommendations
Automatic and Manual transmission sales are relatively balanced.	• *Maintain equal stock of both transmission types* • *Monitor shifts in preference over time*

Q10 Which Models Are Underperforming vs Regional Average?

```
WITH model_sales AS (
SELECT dealer_region, model, COUNT(*) AS total_sales
FROM car_sales GROUP BY dealer_region, model
),
regional_avg_sales AS (
SELECT dealer_region, ROUND(AVG(total_sales),2) AS avg_sales
FROM model_sales GROUP BY dealer_region
)
SELECT m.dealer_region, m.model, m.total_sales,
r.avg_sales, (m.total_sales - r.avg_sales) AS performance_gap
FROM model_sales m
JOIN regional_avg_sales r ON m.dealer_region = r.dealer_region
WHERE m.total_sales < r.avg_sales
ORDER BY performance_gap;
```

■ Insight	■ Recommendations
Some models are dragging down overall regional performance significantly.	• *Reduce or remove stock of underperforming models* • *Replace with regionally popular alternatives*

Q11 Gender Based Purchase Behavior

```
SELECT gender,
COUNT(*) AS total_purchases,
'$ ' || ROUND(SUM(price)/1000000.0, 2) || 'M' AS total_revenue
FROM car_sales
GROUP BY gender
ORDER BY total_revenue DESC;
```

■ Insight	■ Recommendations
Revenue contribution differs noticeably between genders.	• *Design gender-targeted marketing campaigns* • *Stock models preferred by the higher-spending group*

Q12 Most Popular Color by Body Style

```
SELECT body_style, color, COUNT(*) AS total_sales
FROM car_sales
GROUP BY body_style, color
ORDER BY body_style, total_sales DESC;
```

■ Insight	■ Recommendations
Different body styles attract customers with different color preferences.	*• Stock popular colors per body style category* *• Avoid ordering slow-moving color combinations*

■ SECTION 4 — KEY FINDINGS SUMMARY

Area	Key Finding	Action Required
Revenue	Total revenue quantified across all regions	Set annual growth targets
Regions	Clear top & bottom performing regions exist	Redistribute marketing budget
Customers	Income segments show different buying patterns	Personalize offers by segment
Inventory	Regional model preferences vary significantly	Localize inventory strategy
Trends	Monthly growth is inconsistent / seasonal	Run off-season promotions
Dealers	Few dealers outperform the rest	Share best practices network-wide
Products	SUV & Hatchback dominate demand	Increase stock of top styles
Gender	Revenue skewed by gender preferences	Gender-targeted campaigns
Colors	Color-body style combos vary by demand	Optimize color ordering

SQL Skills Demonstrated

■ Aggregations (SUM, AVG, COUNT)	**■ Window Functions (RANK, LAG)**	**■ CTEs (WITH clause)**
■ CASE WHEN Segmentation	**■ Subqueries**	**■ Date Functions (DATE_TRUNC)**
■ JOINs	**■ NULLIF / Error Handling**	**■ CREATE VIEW**